SWAPNIL JOSHI

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PROFESSIONAL SUMMARY

Entry-level Mechanical Engineer with strong mechanical fundamentals and hands-on experience supporting product design, prototyping, testing, and manufacturing process improvement. Comfortable working on the shop floor with operations teams, preparing drawings and work instructions, supporting prototype builds, and analyzing processes for efficiency, cost, and quality improvements. Motivated to grow in a plant-based environment focused on continuous improvement and safety-critical products.

EDUCATION

Clemson University — **August 2023 - May 2025**
Master's, Mechanical Engineering — *GPA: 3.2*

Savitribai Phule Pune University — **July 2018 - June 2022**
Bachelor's, Mechanical Engineering — *GPA: 3.5*

PROFESSIONAL EXPERIENCE

Manufast.in — **Pune, MH, India**
Product Design and Development Engineer — July 2022 - June 2023

- Designed and developed mechanical equipment and fixtures from concept through production release, balancing functional requirements, manufacturability, cost, and ease of assembly in fast-paced manufacturing environments.
- Produced detailed CAD models and engineering drawings for mechanical assemblies, defining tolerances, interfaces, and fabrication details to support repeatable builds and reliable operation on the shop floor.
- Supported prototype-to-production transition by working hands-on with technicians during assembly, debugging fitment issues, improving accessibility, and implementing design changes that increased robustness and reduced assembly time.
- Participated in testing and validation of mechanical assemblies, reviewing test results, identifying root causes of failures, and implementing corrective actions prior to production release.
- Maintained detailed documentation including layouts, drawings, test observations, and engineering changes to support equipment duplication and long-term production use.

Team MechAgrinics — **Pune, MH, India**
R&D and Digging System Head — June 2019 - June 2022

- Led end-to-end development of complex mechanical systems, including concept generation, detailed design, fabrication support, assembly, and validation under real operating conditions.
- Designed machine-like assemblies involving linkages, structural members, hydraulic elements, and actuated mechanisms, balancing strength, stiffness, dynamics, and manufacturability.
- Built and debugged physical prototypes using machining, welding, and manual assembly, iterating designs rapidly based on hands-on testing and observed system behavior.
- Identified opportunities to simplify assemblies, reduce part count, and improve ease of build and maintenance, directly impacting fabrication time and system consistency.
- Worked closely with electrical, fabrication, and operations team members to resolve integration challenges and deliver production-capable hardware under aggressive timelines.
- Gained extensive experience working hands-on with tools, fixtures, and test setups, developing strong intuition for how designs behave once built and operated.

PROJECTS

Suspension and Steering Subsystem Simulation — **Clemson, SC, USA**
Clemson University — January 2025 - May 2025

- Developed a front suspension model and steering linkages in Solidworks; validated through kinematic and load transfer analysis.
- Performed DOE-based sensitivity analysis on suspension geometry, improving cornering stability and ride comfort.

Lightweight Chassis Design and Optimization Study — **Clemson, SC, USA**
Clemson University — August 2024 - December 2024

- Designed a full vehicle frame model and optimized structural members for stiffness-to-weight ratio using topology optimization.
- Conducted FEA and modal analysis to validate safety factors and minimize resonance issues.
- Improved mass efficiency by 14% while meeting all strength and deflection requirements.

Autonomous 360-degree Fire Extinguisher Robot — **Pune, MH, India**
Savitribai Phule Pune University — August 2021 - June 2022

- Designed and built an autonomous 360-degree fire extinguisher robot that maneuvers in various environments to extinguish fires.
- Created a 360-degree fire suppression system within the robot, enabling it to tackle fires from any angle efficiently and reducing emergency response time by 25%.

Hydraulic Lift and Drop Arm — **Pune, MH, India**
Savitribai Phule Pune University — August 2018 - June 2019

- Developed a hydraulic lift and drop arm solution to facilitate smoother operations and move heavy loads, reducing manual handling time by 30% and decreasing injury reports among staff.

SKILLS

Manufacturing Engineering: Process Development, Prototype Assembly & Testing, Work Instructions
CAD & Drawings: Creo, SolidWorks, CATIA V5, GD&T, Manufacturing Drawings
Testing & Validation: Test Setup, Data Collection, Statistical Analysis, Reporting
Tools & Software: Python, JavaScript, SQL,Microsoft Excel, MATLAB (basic), ERP exposure (basic)
Process Improvement: DFM/DFA, Cost Awareness, Workflow Optimization
Materials & Processes: Sheet metal fabrication, welding exposure, tooling coordination